Requests Responses and JSON

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

WORKBOOK · 1 OF 21

Recall and choose your starting point

> Record your starting explanation for Requests Responses and JSON before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Tools and APIs

Without opening the old lesson, explain one key idea from Tools and APIs. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

User Need and AI Contribution

Without opening the old lesson, explain one key idea from User Need and AI Contribution. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Verify Numbers and Dates

Without opening the old lesson, explain one key idea from Verify Numbers and Dates. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 21

Practise with fading help

> Complete the supported Requests Responses and JSON trace and record the first step you can explain without help.

Use W2: Separate failures for **JSON** and **interface contract**s. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 21

Individual reasoning check

> Check the proposed Requests Responses and JSON explanation against the supplied facts before discussing it.

Use the worked case for **JSON** and **interface contract**s. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 21

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Specify independently

Define a **JSON contract** for {"label": "Box", "count": 4}: nonempty text label, integer count 0 through 10. Give a valid case, a wrong-type case and a missing-field case with expected acceptance.

> **My independent explanation, trace and limits**
>
> | |
> |---|
> | |
> | |
> | |
> | |

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 21

Your learning target

> Connect the Requests Responses and JSON target to a trace and a justified conclusion.

Request id=7 returns **status**=200 and {"id":8,"pages":90}. Explain why a successful **status** is insufficient and specify the handling.

WORKBOOK · 6 OF 21

Solve the given case

> Solve the supplied Requests Responses and JSON case and show the deciding evidence.

Request is for B7; **response** is {"id":"B7","pages":120}. Another **response** has malformed **JSON**.

What checks precede use?

Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

WORKBOOK · 7 OF 21

Check a changed case

Apply the procedure to the changed case and record why the conclusion follows.

An API returns {"ok":false,"error":"not_found"}. Your app tries to display **response**.title. What should it do instead?

My independent answer and reason

Write or draw your reasoning and evidence.

WORKBOOK · 8 OF 21

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe **response** before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

WORKBOOK · 9 OF 21

Project checkpoint

> Justify what Requests Responses and JSON adds to Verified Study Buddy and keep the evidence.

Use a stable **request/response contract** between **interface**, retriever, model, and verifier.

The project change and **evidence** I will keep

Write or draw your reasoning and evidence.

WORKBOOK · 10 OF 21

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 21

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

Request id=7 returns **status**=200 and {"id":8,"pages":90}. Explain why a successful **status** is insufficient and specify the handling.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

WORKBOOK · 12 OF 21

W1 Translate a value

Translate a value. Apply the supplied Requests Responses and JSON facts and keep your reasoning.

What Python value does **JSON** null become? What **JSON** value represents Python True?

My reasoning and evidence

WORKBOOK · 13 OF 21

W2 Separate failures

> Separate failures. Apply the supplied Requests Responses and JSON facts and keep your reasoning.

Classify {"minutes": "5"} and {minutes: 5} as valid **JSON** text or malformed **JSON**.

My reasoning and evidence

W3 Repair the interface

> Repair the interface. Apply the supplied Requests Responses and JSON facts and keep your reasoning.

A parser accepts a **JSON** list then crashes on .get. What check belongs before field access?

My reasoning and evidence

WORKBOOK · 15 OF 21

W4 Specify independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Specify independently. Apply the supplied Requests Responses and JSON facts and keep your reasoning.

Define a **JSON contract** for {"label": "Box", "count": 4}: nonempty text label, integer count 0 through 10. Give a valid case, a wrong-type case and a missing-field case with expected acceptance.

My reasoning and evidence

WORKBOOK · 16 OF 21

W5 Limit parsing

> Limit parsing. Apply the supplied Requests Responses and JSON facts and keep your reasoning.

A **JSON** record parses successfully. Does that prove its **observation** is correct?

My reasoning and evidence

WORKBOOK · 17 OF 21

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 18 OF 21

Pictorial case practice: Requests Responses and JSON

> A lunch-count response must satisfy a declared JSON contract. Explain the labelled step

> before checking the changed condition.

A lunch-count **response** must satisfy a declared **JSON contract**.

The toy **response** has exactly class_code text and portions integer 0..30. Example: {"class_code":"R8","portions": 12}.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-3/22

Requests Responses and JSON

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Read fields

class_code is nonempty text; portions is the integer 12.



2. Check range

0 <= 12 <= 30, so the declared count is within range.



3. Accept structure

Accept this response shape; acceptance does not prove that 12 portions were actually counted.

CHANGE ONE THING

Replace 12 with the text "12".

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Would converting the text silently prove the source count is correct? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Verified Study Buddy record.

Project: Verified Study Buddy

Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	

Evidence to keep	My record
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 20 OF 21

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A lunch-count **response** must satisfy a declared **JSON contract**.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Accept this **response** shape; acceptance does not prove that 12 portions were actually counted.

1. Accept structure

2. Read fields

3. Check range

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Accept structure Accept this **response** shape; acceptance does not prove that 12 portions were actually counted. Read the picture in order.

Accept this **response** shape; acceptance does not prove that 12 portions were actually counted. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Replace 12 with the text "12". Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. We can decide the result without examining the changed **input** or the procedure.

3. Reject the **response** under this strict **contract**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Reject the **response** under this

> strict **contract**. Digits inside quotes represent text, whereas portions requires an integer.

Repair a missing building block

Review Tools and APIs (ai-3/21). Goal: Explain the **mechanism** and terminology of tools and api **contract**s. Short practice: Does a valid C7 ID permit the extra lending action?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review User Need and AI Contribution (ai-3/19). Goal: Explain the **mechanism** and terminology of capstone requirements and corpus. Short practice: What non-AI baseline would reveal whether this helper adds value?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Verify Numbers and Dates (ai-3/16). Goal: Explain the **mechanism** and terminology of claim support and citation evaluation.

Short practice: Which part can be corrected immediately from existing **evidence**, and which needs follow-up?. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

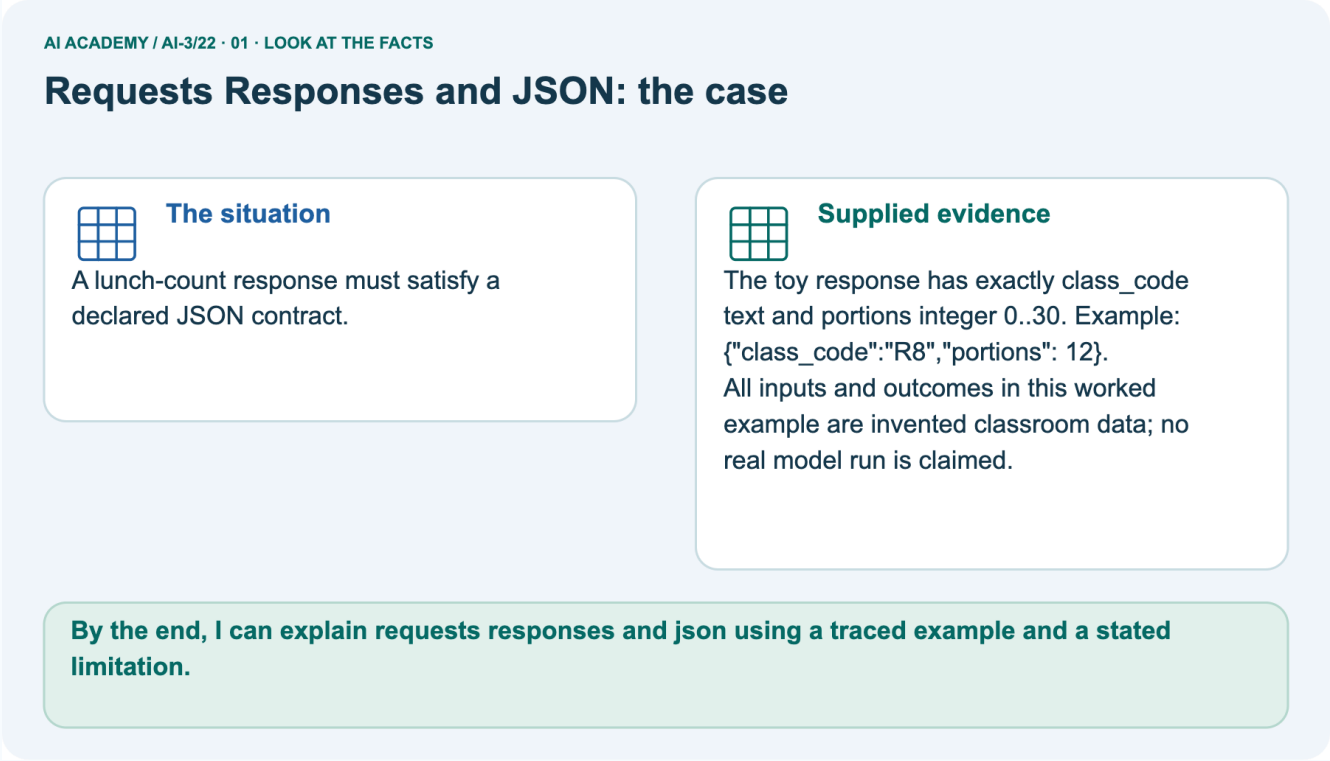
What I repaired, the evidence I kept and the next test

WORKBOOK · 21 OF 21

Visual concept journey

Explain each Requests Responses and JSON picture from its labels, then record a relevant limit.

01. Requests Responses and JSON: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

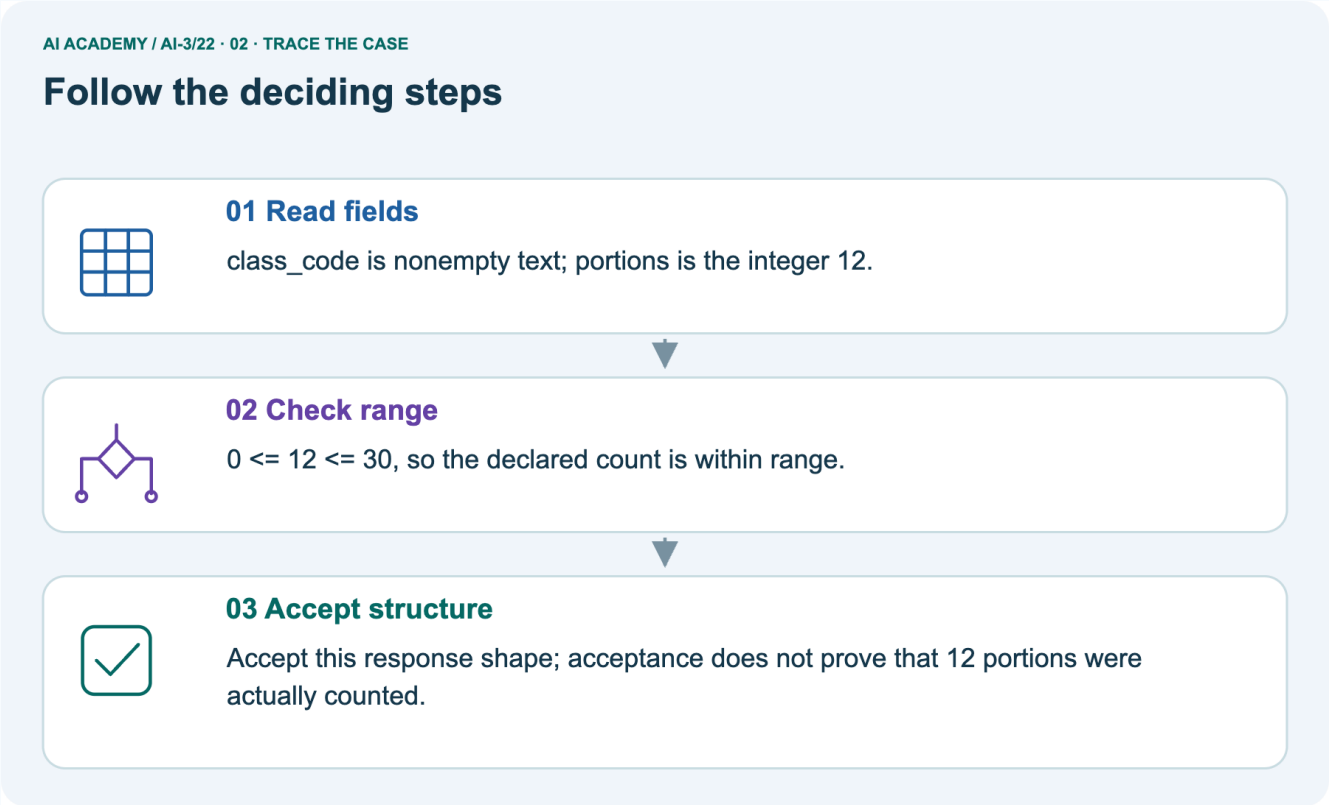
A lunch-count **response** must satisfy a declared **JSON contract**. Supplied facts: The toy **response** has exactly class_code text and portions integer 0..30. Example: {"class_code":"R8","portions": 12}.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



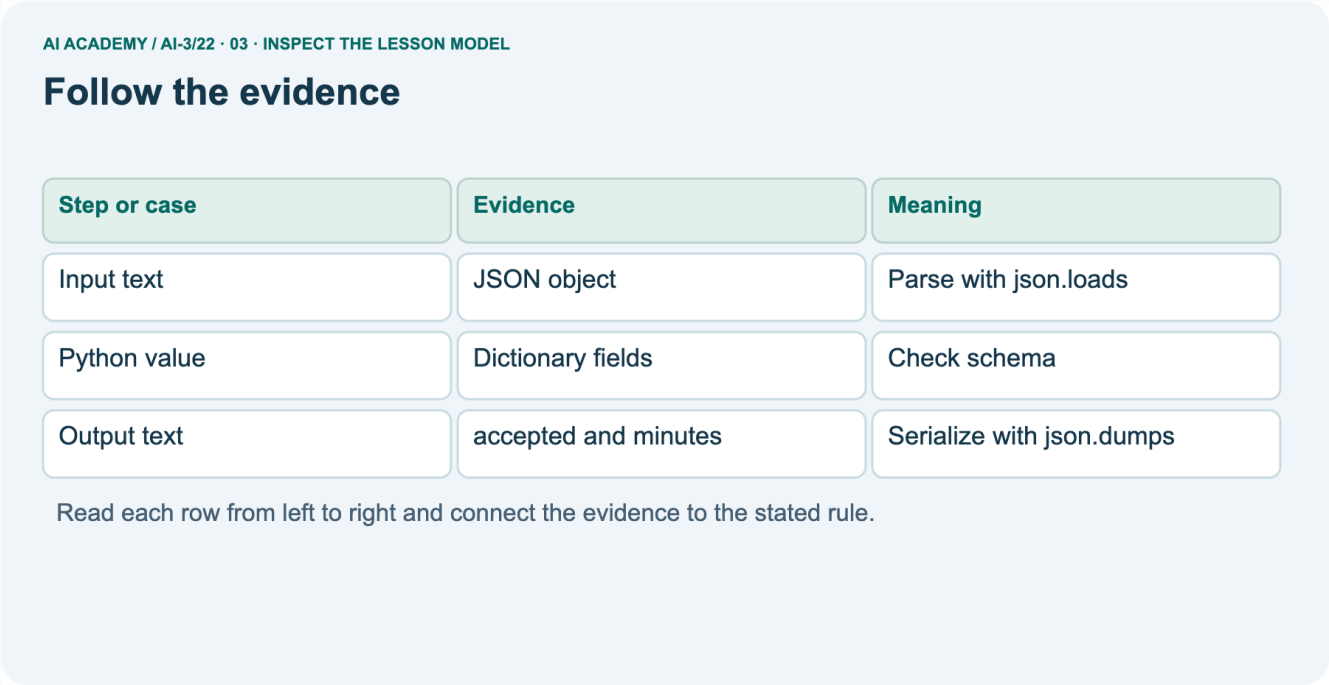
Follow the labelled picture with the matching explanation. Open the vector diagram

Read fields: class_code is nonempty text; portions is the integer 12. → Check range: 0 <= 12 <= 30, so the declared count is within range. → Accept structure: Accept this **response** shape; acceptance does not prove that 12 portions were actually counted.

> Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Follow the evidence



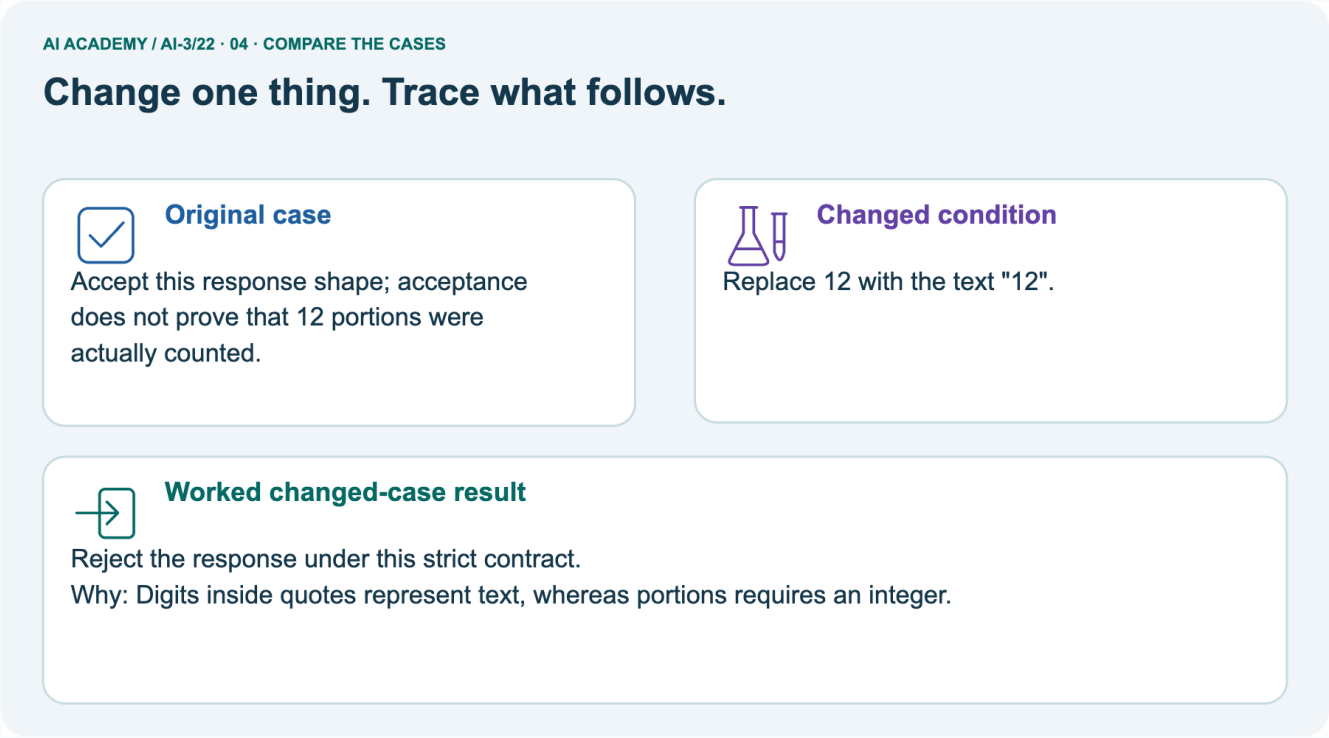
Follow the labelled picture with the matching explanation. Open the vector diagram

Follow the **evidence**. **Input** text | **JSON object** | Parse with **json**.loads. Python value | Dictionary fields | Check schema. **Output** text | accepted and minutes | Serialize with **json**.dumps

> Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

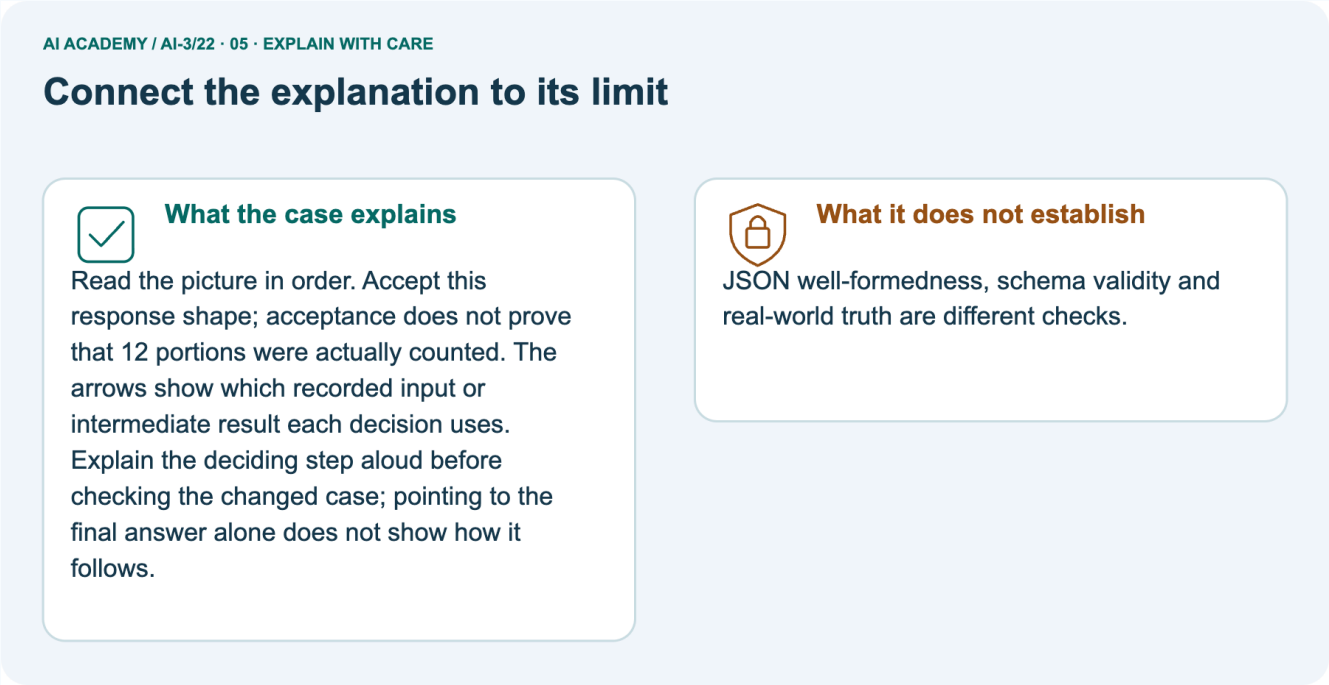
Original case: Accept this **response** shape; acceptance does not prove that 12 portions were actually counted. Changed condition: Replace 12 with the text "12". Expected result: Reject the **response** under this strict **contract**.

Reason: Digits inside quotes represent text, whereas portions requires an integer.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

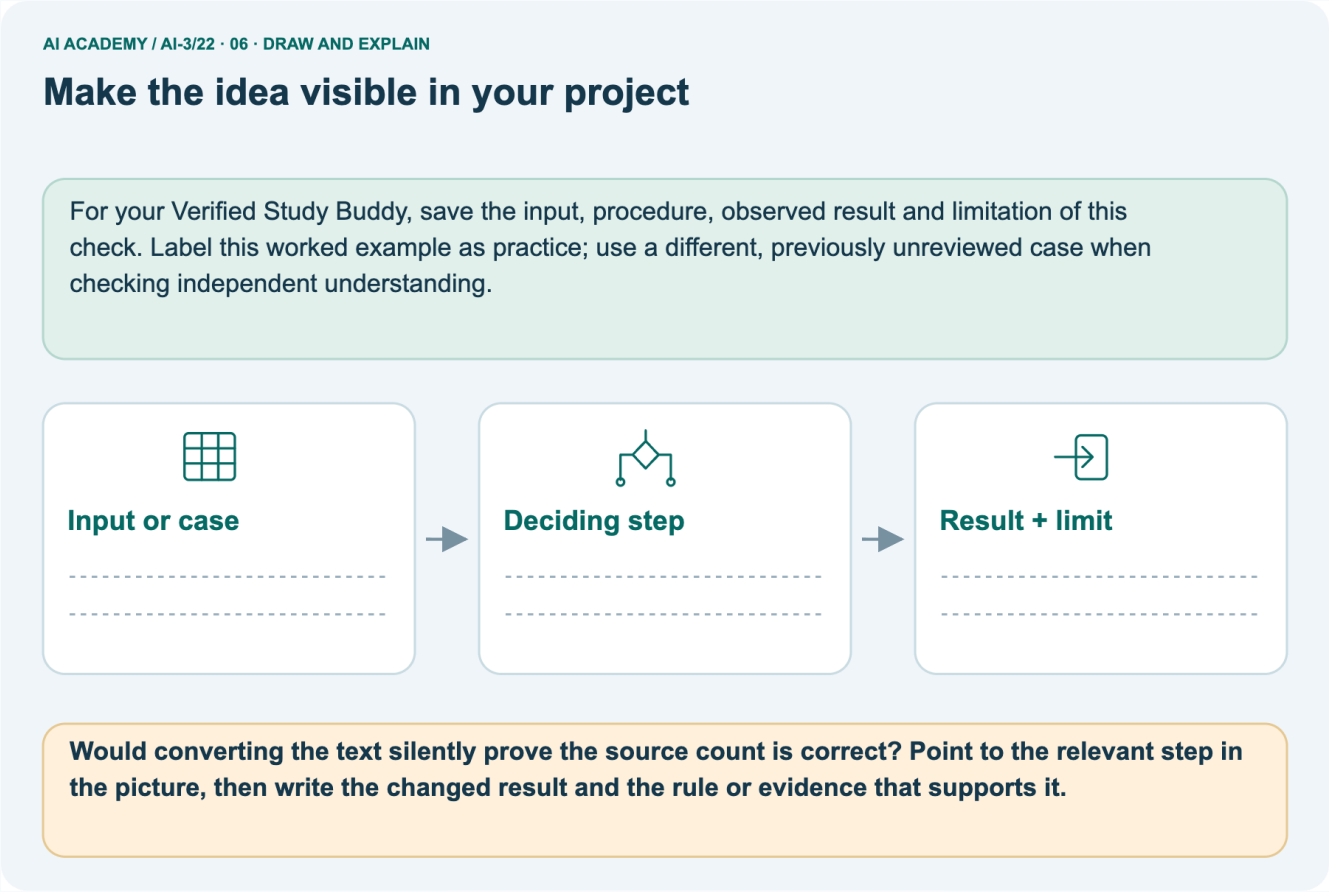
Read the picture in order. Accept this **response** shape; acceptance does not prove that 12 portions were actually counted. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows. Limit: **JSON** well-formedness, schema validity and real-world truth are different checks.

> Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Would converting the text silently prove the source count is correct? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.